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Project Title:	AC heat capacity measurement using gravity gradient approach		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
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Zip Code:	91109	**Congressional District:**	29
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	This proposed ground-based experiment will measure the specific heat at constant volume C_V of He-3 fluid in the asymptotic region of the liquid-vapor critical point. Our goal is to experimentally determine the critical amplitudes of the specific heat. Knowing the leading critical amplitudes will enable us to accurately determine the crossover parameters, namely the Wegner expansion amplitudes. We will compare our experimental ratios of the leading amplitudes and the Wegner expansion amplitudes to the recently improved Renormalization Group theory predictions for the universal values. The strong divergence of the compressibility leads to a strong density stratification across a fluid layer in a gravitational field. Consequently, the predicted divergences in thermodynamic properties near liquid-vapor critical points are smeared out. Because of this gravity induced smearing, the liquid-vapor critical point is ideally suited for experimental investigation in a microgravity environment. The only previous microgravity experiment for static properties near the		

Task Description:	liquid-vapor critical point is the measurement of the specific heat at constant volume, C_V, along the critical isochore of sulfur hexafluoride, SF-6. That experiment was performed during the German Spacelab Mission D-2 on board of a Space Shuttle flight. However, the crossover analysis of the SF-6 C_V data shows results that are not consistent with recent RG theoretical predictions. This indicates the need of making further measurements of C_V near the liquid-vapor critical point under the microgravity condition. We propose to measure the He-3 specific heat in the asymptotic region on the ground using a gravity cancellation approach. DC heat will be applied at the bottom of a He-3 fluid layer to generate a steady state condition without any internal mass motion. The upward thermal expansion induced by the resultant temperature gradient can compensate the downward hydrostatic compression induced by the fluid's own weight. Consequently, the gravity-induced density stratification can be significantly reduced by a factor of 100. The optimum gravity cancellation condition occurs at the onset of convection. The convection onset temperature difference is primarily determined by the buoyancy force required to overcome the hydrostatic compression. The measurement of the convection onset is actually the measurement of the cancellation of the density stratification. At the convection onset, the He-3 fluid layer can be considered homogeneous in density in a laboratory experiment over the reduced temperature range 10^{-6} to 10^{-4}. Specific heat measurements can therefore be performed on the ground over two decades in the asymptotic region along the critical isochore. For a 0.05 cm thick He-3 fluid layer, the onset temperature difference will be Delta T_{ad} = 1.72 micro K and it is essentially a constant in the asymptotic region. This small temperature difference will cause a systematic deviation of less than 0.1% in the measured C_V. The specific heat will be measured using an AC technique. Because there is a tremendous difference in the time scale between the critical slowing-down of the density relaxation and the critical speeding-up of the temperature equilibration, the reduced density stratification by DC heating will not respond to fast AC heating. Hence, a steady-state AC calorimetry will be used to measure the fluid's heat capacity. Even though the critical DC heating already drives the fluid layer to the edge of convection, the additional AC heating will not cause convection because convection takes time to ensue and because the corrected local Rayleigh number can be adjusted to be less than the critical Rayleigh number. Thus, AC specific heat measurements can be taken under these stead-state conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
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